

Differential Effect of Paradoxical Sleep Deprivation on Acquisition and Retrieval of Conditioned Taste Aversion in Rats

H. VENKATAKRISHNA-BHATT¹, J. BURES AND O. BURESOVA

Institute of Physiology, Czechoslovak Academy of Sciences, Prague, Czechoslovakia

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VENKATAKRISHNA-BHATT, H., J. BURES AND O. BURESOVA. *Differential effect of paradoxical sleep deprivation on acquisition and retrieval of conditioned taste aversion in rats*. *PHYSIOL. BEHAV.* 20(2) 101–107, 1978. — Paradoxical sleep deprivation (PSD) was used to interfere with acquisition or retrieval of conditioned taste aversion (CTA). PSD was achieved by confining rats to small pedestals placed on an electrified grid floor. Fifteen-min access to 0.1% sodium saccharin (conditioned stimulus-CS) by water deprived rats was followed 30–120 min later by intraperitoneal injection of lithium chloride (unconditioned stimulus-US, 0.15 M, 2% to 4% body weight). Retention was tested 1 to 5 days later by offering the animal saccharin again. A 24-hr PSD preceding saccharin drinking prevented CTA acquisition. CTA disruption was diminished by a 2-hr, and completely abolished by an 8-hr interval of home cage recovery inserted between the 24-hr PSD and saccharin presentation. CTA was slightly facilitated by 2-hr PSD in the CS-US interval. The 24-hr PSD preceding CS caused the same CTA disruption when followed by free sleep opportunity or by continued PSD in the 2-hr CS-US interval. Twenty-four-hr PSD preceding retention testing slightly improved retrieval of well established CTA. It is concluded that PSD interferes with the formation of the short-term gustatory trace of CTA but does not affect retrieval of CTA engrams. Gradual compensation for the PSD effect on CTA learning by pre-acquisition sleep suggests that the processes responsible for CTA disruption and recovery correspond to depletion and repletion of the same neurotransmitter stores.

Sleep deprivation Taste aversion Paradoxical sleep deprivation Conditioned taste aversion

CONDITIONED taste aversion (CTA) has been employed recently to investigate various components of the memory forming process. The long delay between the gustatory stimulus (CS) and the visceral symptoms of poisoning (US) makes it possible to isolate at least 3 phases of CTA acquisition, i.e., formation of the short-term gustatory trace, persistence of this trace, and its association with poisoning, which are differentially sensitive to interfering treatments such as cortical spreading depression [4, 5, 9], anaesthesia [7,36], hypothermia [19], electroconvulsive shock [22, 23, 24, 31], brain anoxia [18]. The results of these studies indicate that the last two phases are extremely resistant to severe impairment of brain function. The conditioned taste aversion acquisition was not prevented by bilateral functional decortication, anaesthesia, hypothermia and cerebral anoxia induced during the CS-US interval and maintained over the most severe stage of poisoning. On the other hand, the above procedures prevented CTA acquisition when applied immediately before presentation of the gustatory CS. Processing of the taste signals and formation of the short-term gustatory trace seem to be the most fragile links in the series of events leading to the establishment of permanent CTA engrams.

Interpretation of the above findings was complicated by the fact that in absence of spontaneous consummatory behaviour, gustatory stimuli had to be presented by forced feeding or by intraperitoneal injection of high concentrations of the flavoured substance. Although taste stimuli applied in this way can serve as CS in intact animals [3,10], it was desirable to add to the list of CTA impairing treatments a procedure which does not interfere with spontaneous feeding and drinking.

Danguir and Nicolaidis [8] recently reported that paradoxical sleep deprivation (PSD) prevents acquisition of CTA to the taste of lithium chloride. The PSD treatment was considerably more effective when applied before than after presentation of the gustatory CS (0.15 M LiCl) which also served in this case as the US. The impairment was a function of PSD duration: it appeared rather abruptly after 7.5 hours of pre-CS PSD and increased only little with prolongation of the pre-CS PSD treatment. On the contrary, 4 days of PSD after lithium chloride drinking caused only a mild reduction of CTA. In contrast with the studies describing disruption of learning by post-trial PSD [1, 12, 34] and increased incidence of paradoxical sleep after learning [26, 29, 38], the results stress the effect of PSD on

¹ Visiting scientist from the Division of Medical and Industrial Toxicology, National Institute of Occupational Health, Meghani Nagar, Ahmedabad 380 016, (Gujarat) India.

the initial processing of gustatory signals. The purpose of the present study was to investigate the PSD effect with a different CTA paradigm and to compare it with the other CTA disrupting procedures.

GENERAL METHOD

Animals

The experiments were performed in male hooded rats (Druckrey strain) aged 2–3 months. The animals were housed in group cages (7 animals/cage) with free access to standard laboratory diet pellets. Temperature of the animal room was kept constant and illumination corresponded to the natural lighting cycle. The rats were maintained on a 23.75-hr water deprivation schedule with water available only in the testing apparatus for 15 min daily.

Apparatus

The drinking box consisted of a plastic enclosure (49 × 15 × 30 cm) with a grid floor and a calibrated pipette attached to the narrow front wall. The drinking spout was 6 cm above the floor level. The liquids consumed were measured with 0.1 ml accuracy.

Procedure

The water deprived rats were given 15-min access to water on Days 1 and 2. On Day 3, water was substituted by 0.1% sodium saccharin, which was followed 30 or 120 min later by intraperitoneal injection of 0.15 M lithium chloride (2 or 4% body weight). In rats receiving the lower LiCl dose, retention was tested on Day 4. After the higher LiCl dose rats had free access to water on Days 4, 5 and 6. The water deprivation schedule was resumed on Days 7 and 8, and the retention test was postponed to Day 9. During retention testing the animal was offered saccharin and the volume of the fluid consumed was measured.

Paradoxical Sleep Deprivation

The pedestal technique proposed by Morden *et al.* [30] was adapted for the purposes of the CTA experiments. Placing the platforms into water tanks was impractical, since the availability of water made water deprivation unreliable. The pedestal (plastic cylinder 8 cm high and 7 cm in diameter) was placed into the centre of a 20 × 20 × 40 cm compartment with electrifiable grid floor. Eight to 16 animals were simultaneously treated in compartments connected to the same shock source (2 mA, 50 Hz, 0.5 sec

on, 3 sec off). Rats were adapted to sitting on the platform by being placed for 2 to 3 hr onto 80 cm high pillars with circular upper surface (7 cm in diameter). After each PSD treatment the animals were briefly offered food in the home cage.

EXPERIMENT 1

The purpose of the first series of experiments was to replicate the basic finding of Danguir and Nicolaidis [8] using saccharin as the CS and LiCl as the US. The procedure described in GENERAL METHOD was employed with a few modifications. After conclusion of drinking on Day 2, the experimental animals (Group B₁) were placed on the pedestals and maintained there for 23 hr. Then they were allowed to eat for 30 min in the home cage before being offered access to saccharin in the drinking box. The rats were poisoned (0.15 M LiCl, 4% body weight) 45 min after onset of saccharin drinking. Retention was tested 6 days later. The control animals (Group A₁) received no PSD treatment.

Results

Water and saccharin consumption during various phases of Experiment 1 is shown in Table 1 (Groups A₁ and B₁). The experimental and control groups were matched according to water intake on Days 1 and 2. The sleep deprived animals consumed significantly less saccharin during the 15-min access to the drinking spout on Day 3, but this finding could not be replicated in Experiment 4. The retention test revealed marked saccharin aversion in the control Group A₁ ($df = 11$, $t = 12.28$, $p < 0.001$, paired comparison), whereas in the experimental Group B₁ saccharin consumption was undiminished ($df = 15$, $t = 0.4$, NS, paired comparison).

These results indicate that the disruption of CTA acquisition by 24-hr PSD preceding saccharin drinking is not limited to the experimental situation used by Danguir and Nicolaidis [8] but also applies to another standard CTA paradigm employing a different gustatory stimulus and a longer CS-US delay.

EXPERIMENT 2

The striking effect of the 24-hr PSD suggests that the failure to acquire CTA is due to brain depletion of mediators synthesized during REM sleep or to other consequences of neural fatigue and exhaustion. In this case restitution of normal sleep cycle should gradually restore

TABLE 1
WATER AND SACCHARIN CONSUMPTION DURING HABITUATION, LEARNING AND
CTA RETRIEVAL TESTING IN EXPERIMENTS 1 AND 2

Gr	n	Day 2 (water)	Day 3 (saccharin)	Day 8 (water)	Day 9 (saccharin)	(D ₉ /D ₂)·100
A ₁	12	11.4 ± 0.3	11.1 ± 0.3	11.5 ± 0.4	4.4 ± 0.6	38.6 ± 5%
B ₁	16	9.9 ± 0.5	7.7 ± 0.7	11.3 ± 0.6	9.7 ± 0.5	98.0 ± 5%
C ₁	8	11.7 ± 1.0	10.8 ± 0.8	12.1 ± 0.9	8.3 ± 1.0	70.9 ± 8%
D ₁	8	11.9 ± 0.9	9.8 ± 0.9	12.9 ± 0.8	8.4 ± 1.2	70.5 ± 10%
E ₁	16	10.5 ± 0.6	11.4 ± 0.7	11.0 ± 0.7	4.8 ± 0.7	45.7 ± 6%
F ₁	7	11.3 ± 1.3	12.5 ± 1.1	12.7 ± 1.0	5.8 ± 1.0	51.3 ± 8%
G ₁	8	10.1 ± 0.4	11.8 ± 0.7	11.7 ± 0.8	5.1 ± 0.6	50.5 ± 6%

the deficit and reinstate CTA learning. The purpose of Experiment 2 was to test this hypothesis by allowing the rats to spend various intervals between the 24-hr PSD and CTA training in the home cage. The procedure described in GENERAL METHOD and Experiment 1 was modified only in timing the onset of the PSD treatment on Day 2 which preceded the drinking test for 2 to 24 hr. Consequently, PSD terminated 2–24 hr before saccharin presentation on Day 3.

Results

Table 1 summarizes the results obtained in Groups C_1 , D_1 , E_1 , F_1 , G_1 tested with the recovery times of 2, 4, 8, 16 and 24-hr respectively. There were no significant differences between these groups in water consumption on Days 1 and 2 and in saccharin intake on Day 3. Reduction of drinking seen during the first 15 min after PSD in Experiment 1 was not observed in rats allowed at least 2 hr of post-PSD recovery. Saccharin consumption during the retention test was a function of the duration of post-PSD recovery. One-way analysis of variance employed to test the differences between all pairs of retrieval data (last column of Table 1) showed that Groups C_1 and D_1 were significantly different from Group A_1 ($F(6,68) = 4.59$, $p < 0.05$) and ($F(5,68) = 4.44$, $p < 0.05$, respectively) as well as from Group B_1 ($F(2,68) = 3.87$, $p < 0.01$ and $F(3,68) = 4.02$, $p < 0.05$) but not from the other groups. Group B_1 was also significantly different from the groups given a longer post-PSD recovery (B_1 versus F_1 ; $F(4,68) = 6.74$, $p < 0.01$; B_1 versus G_1 ; $F(5,68) = 6.89$, $p < 0.01$; B_1 versus E_1 ; $F(6,68) = 7.46$, $p < 0.01$). Other differences between means did not reach the level of significance. Essentially similar results were obtained when the saccharin consumption during the retention test was expressed in percentages of water consumption on Day 2 or related to body weight. Figure 1 graphically represents the data derived from this experiment.

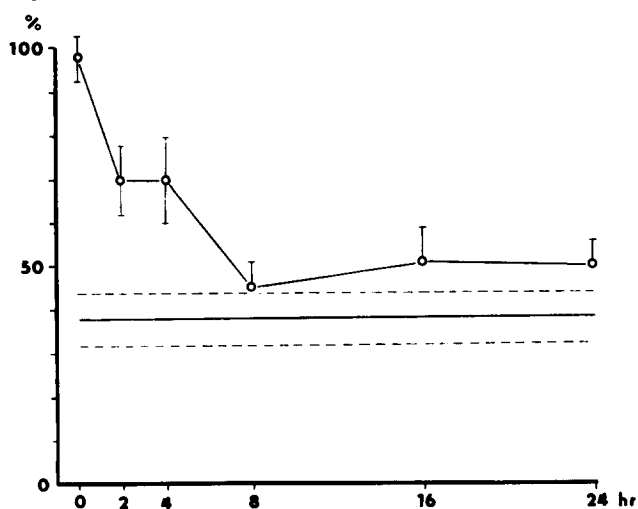


FIG. 1. The effect of 24-hr PSD followed by various intervals of recovery on CTA acquisition. Abscissa: time between PSD termination and saccharin presentation. Ordinate: Saccharin intake during the retention test expressed in percentages of saccharin consumption during the acquisition session. The horizontal line corresponds to CTA retrieval in control animals not exposed to PSD. Vertical lines denote the SEM values.

The above results indicate that even 2 and 4 hours of sleep opportunity partly compensate for the effect of 24-hr PSD on CTA acquisition and that after 8 hr of home cage relaxation CTA learning approaches that achieved by undeprived control rats (Fig. 1). Danguir and Nicolaidis [8] found that CTA acquisition is disrupted by CSD lasting 7.5 hr but not 5.0 hr. Similar dynamics of the processes leading to CTA disruption and CTA recovery after PSD suggest that they correspond to depletion and repletion of the same neurotransmitter stores.

EXPERIMENT 3

The disruption of CTA acquisition by PSD preceding saccharin drinking can be due to interference with the formation of the short-term gustatory trace, with the persistence of this trace or with the CS-US association and subsequent consolidation of the permanent CTA engram. Danguir and Nicolaidis [8] found that CTA is little affected by prolonged (96-hr) posttrial PSD. Experiment 3 tested the effect of PSD on the persistence of the short-term gustatory trace. The procedure described in GENERAL METHOD was employed with the CS-US interval of 2 hr. The isotonic LiCl solution was applied in a volume corresponding to 2% body weight in Groups A_2 and B_2 and 4% body weight in Groups C_2 and D_2 . Paradoxical sleep deprivation was achieved by placing the rats on tall pillars (80 cm above the floor level) during the 2-hr CS-US interval (Group B_2 and D_2). The control groups A_2 and C_2 spent the CS-US interval in their home cages. Retention was tested on Day 4 in groups A_2 and B_2 and on Day 9 in groups C_2 and D_2 .

Results

Table 2 shows consumption of saccharin and water in all 4 groups. Although the groups were matched according to the water intake on Day 1, Group B_2 drank significantly less saccharin on Day 3 than the Group A_2 . Higher fluid consumption on groups C_2 and D_2 is due to their higher body weight. Poisoning was too weak in Group A_2 in which saccharin consumption was not significantly lower during retention testing than on Day 3. ($df = 9$, $t = 1.21$, NS, paired comparison). Paradoxical sleep deprivation treatment in the CS-US interval facilitated CTA in Group B_2 which drank significantly less saccharin during retention testing than during acquisition ($df = 22$, $t = 4.5$, $p < 0.001$, paired comparison). Stronger poisoning caused significant saccharin aversion in groups C_2 ($df = 9$, $t = 15.6$, $p < 0.001$, paired comparison) and D_2 ($df = 13$, $t = 20.2$, $p < 0.001$, paired comparison). A two-way analysis of variance of the retrieval data in Table 2 shows significant main effects of both lithium chloride concentration ($F(1,53) = 60.12$, $p < 0.01$) and PSD ($F(1,53) = 9.4$, $p < 0.01$), but no measurable interaction of the above factors ($F(1,53) = 0.02$, NS).

The fact that PSD in the CS-US interval does not impair CTA acquisition is not unexpected, since PSD of this duration is ineffective even when applied before the gustatory stimulus [8]. Facilitation of CTA acquisition in groups B_2 and D_2 suggests, however, that post-CS PSD affects CTA in a different way than pre-CS PSD. Hennessey *et al.* [16] recently reported enhanced CTA acquisition in rats pre-treated with ACTH. The 2-hr stay on the pillar is a

TABLE 2
CONSUMPTION OF WATER AND SACCHARIN IN EXPERIMENT 3

Gr	n	Day 2 (water)	Day 3 (saccharin- Learning)	Day 4 (saccharin- Retention)	Day 9 (saccharin- Retention)	(R/L)·100
A ₂	10	10.7 ± 0.6	10.6 ± 0.6	9.7 ± 0.8	—	91.5 ± 7%
B ₂	23	9.1 ± 0.3	7.9 ± 0.3	6.0 ± 0.3	—	75.0 ± 4%
C ₂	10	12.0 ± 0.6	11.6 ± 0.5	—	6.0 ± 0.4	53.3 ± 3%
D ₂	14	12.1 ± 0.5	11.7 ± 0.4	—	4.6 ± 0.4	39.3 ± 3%

stressful situation which may cause considerable pituitary-adrenal activation and thus increase the effectiveness of subsequent association.

EXPERIMENT 4

The question whether the 24-hr PSD preceding saccharin drinking in Experiment 1 prevented CTA acquisition by disrupting the formation of the short-term gustatory trace or by impairing the gustatory trace-poisoning association can be answered by combination of the approaches employed in Experiments 1 to 3. If the PSD effect on the gustatory trace-poisoning association is an important factor, then impairment of CTA acquisition should be stronger with 24-hr PSD treatment continuing during the CS-US interval, then with PSD limited to the pre-CS period. The difference between these treatments should be proportional to the duration of the CS-US interval, because free sleep in this period may partly compensate for the putative PSD effects on the gustatory trace-poisoning association. Experiment 4 tested the above assumption. The procedure described in GENERAL METHOD was employed with CS-US interval of two hours. In the experimental groups B₃ and C₃, PSD started after conclusion of water drinking on Day 2 and was maintained for 23 hr. The rats were then offered food (30 min) and saccharin (15 min). Group B₃ spent the 2-hr CS-US interval in home cages, Group C₃ was returned to the pedestals. The control animals (Group A₃) received no PSD treatment. Lithium chloride (0.15 M, 4% body weight) was applied 2 hr after conclusion of saccharin drinking and retention was tested on Day 9.

Results

Table 3 summarizes the water and saccharin intake data of Experiment 4. All three groups were matched according to water consumption on Day 2. Whereas the control Group A₃ showed significant CTA ($df = 12$, $t = 16.7$, $p < 0.001$, paired comparison), saccharin consumption during the retention test was not significantly different from

the water consumption on Day 2 in Group C₃ ($df = 6$, $t = 1.8$, $p < 0.05$, paired comparison) and was even increased in Group B₃ ($df = 7$, $t = 4.5$, $p < 0.01$, paired comparison). One-way analysis of variance applied to the retrieval data of Table 3 showed that the control group A₃ was significantly different from groups B₃ ($F(3,25) = 14.6$, $p < 0.01$) and C₃ ($F(2,25) = 12.2$, $p < 0.01$), but that there was no difference between groups B₃ and C₃ ($F(2,25) = 2.4$, NS).

The above results support the view that PSD interferes with the formation of the short-term gustatory trace but they do not exclude the possible effect of PSD on the gustatory trace-poisoning association. If such effects exist they cannot be detected when the gustatory trace is practically absent.

EXPERIMENT 5

Treatments interfering with CTA acquisition usually fail to prevent CTA retrieval. Thus bilateral functional decortication and anaesthesia do not block CTA extinction induced by forced feeding of saccharin [6] or by intraperitoneal injection of saccharin [7]. Experiment 5 tested the effect of the 24-hr PSD on CTA retrieval.

The procedure described in GENERAL METHOD was slightly modified. Saccharin drinking followed 30 min later by LiCl poisoning (0.15 M, 4% body weight) was applied on Day 3 in groups A₄ and B₄ and on Days 3 and 5 in groups C₄ and D₄. Since groups C₄ and D₄ displayed on Day 5 marked CTA to saccharin, the small amount of spontaneously swallowed saccharin was supplemented by additional intraperitoneal injection of 2% saccharin solution (1% body weight). Retention was tested on Day 9 in groups A₄ and B₄ and on Day 11 in groups C₄ and D₄. Groups B₄ and D₄ were subjected to PSD during the 24 hr preceding the retention test.

Results

The four groups were matched according to water consumption on Day 2 and saccharin consumption on Day

TABLE 3
CONSUMPTION OF WATER AND SACCHARIN IN EXPERIMENT 4

Gr	n	Day 2 (water)	Day 3 (saccharin- Learning)	Day 8 (water)	Day 9 (saccharin- Retention)	(D ₉ /D ₂)·100
A ₃	13	11.6 ± 0.7	13.0 ± 0.6	14.6 ± 0.8	6.5 ± 0.4	50.3 ± 3%
B ₃	8	11.2 ± 0.7	14.8 ± 0.5	16.3 ± 0.8	14.3 ± 0.7	127.6 ± 6%
C ₃	7	10.2 ± 0.7	9.8 ± 0.8	11.0 ± 0.5	11.7 ± 0.9	114.7 ± 8%

TABLE 4
CONSUMPTION OF SACCHARIN AND WATER IN EXPERIMENT 5

Gr	n	Day 2 (water)	Day 3 (saccharin- Learning)	Day 8 (water)	Day 9 (saccharin- Retention)	(D ₉ /D ₃)·100	
A ₁	12	11.4 ± 0.3	11.1 ± 0.3	11.5 ± 0.4	4.4 ± 0.6	39.6 ± 6%	
B ₁	10	11.0 ± 0.6	11.7 ± 0.4	11.0 ± 0.4	2.4 ± 0.7	20.5 ± 6%	

Gr	n	Day 2 (water)	Day 3 (saccharin- Learning ₁)	Day 5 (saccharin- Learning ₂)	Day 10 (water)	Day 11 (saccharin- Retention)	(R/L)·100
C ₁	10	10.4 ± 0.5	11.6 ± 0.6	1.8 ± 0.1	10.8 ± 0.7	2.8 ± 0.5	24.1 ± 4%
D ₁	10	10.8 ± 0.7	11.5 ± 0.6	1.9 ± 0.1	10.2 ± 0.6	1.3 ± 0.3	11.3 ± 2%

3 (Table 4). Groups C₁ and D₁ showed a marked decrease of saccharin intake during the second acquisition session (Day 5). A two-way analysis of variance of the retrieval data in Table 4 shows significant main effect of both level of training ($F(1,38) = 5.50, p < 0.05$) and of PSD treatment ($F(1,38) = 10.43, p < 0.01$) but no significant interaction of these factors ($F(1,38) = 0.05, NS$). The differential effect of PSD on CTA acquisition and retrieval resembles the results obtained with other agents such as bilateral cortical spreading depression and anaesthesia.

DISCUSSION

Differential effect of PSD on CTA acquisition and retrieval supports the view [6,7] that these processes are subserved by different neural mechanisms. Whereas validity of the evidence based on the effects of cortical spreading depression and anaesthesia is somewhat limited by the fact that these interventions prevent spontaneous feeding and drinking, so that the gustatory signals are processed under unusual conditions (forced feeding, intraperitoneal injection), PSD does not interfere with consummatory behaviour and represents therefore a more specific method of CTA blockade. Unfortunately the long duration of effective PSD complicates the interpretation of results.

The experimental evidence is quite conclusive on one point: consolidation of CTA engram is not directly related to paradoxical sleep. It is almost unaffected by 96 hr of postacquisition PSD [8] and the disruption demonstrated in Experiment 1 is not alleviated by unlimited postacquisition sleep. In this respect CTA appears to differ from the discrimination learning and shuttle box avoidance learning, the consolidation of which is disrupted by post-trial PSD [21, 27, 33, 34]. On the other hand, Fishbein and Gutwein [14] reported that 3-hr PSD rather facilitates than disrupts consolidation of a passive avoidance task. The available evidence does not preclude the possibility of retarded CTA memory consolidation during maintained postacquisition PSD. Such effects were reported by Fishbein [12] who found blockade of passive avoidance in retention tests performed 30 min and 3-hr but not 24 hr after 3 to 5 days of posttrial PSD.

It seems that CTA acquisition requires normal function of a system, the activity of which is gradually disrupted by PSD and gradually restored by REM sleep opportunity. The

metabolic nature of the underlying processes is suggested by the slow dynamics of deprivation and recovery effects. Stern and Morgane [40] recently reviewed the experimental evidence indicating that PSD blocks the activity of catecholaminergic neurons projecting from the lateral reticular formation of medulla, pons, and midbrain to hypothalamus, limbic system, hippocampus and neocortex [2,41]. Intense activity of this system during wakefulness gradually induces synaptic fatigue due to decreased availability of catecholamine transmitters. This state generates the need for REM sleep which restores the synaptic availability of catecholamines in a homeostatic fashion. Regulation of catecholamine level is disrupted by PSD which eventually leads to functional blockade of catecholaminergic transmission. One of the putative behavioural consequences is the impairment of learning observed after PSD [14, 15, 17].

Design of many PSD experiments precludes answering the question whether PSD interferes with acquisition, consolidation, or retention of memory traces. There are only few studies addressing this issue more specifically. Stern [39] suggested that PSD impairs short-term memory since he found good retrieval of a passive avoidance reaction immediately (5 sec) after acquisition, but significant deterioration of retrieval 5 min after learning. On the other hand, Fishbein [11] found that 3 days of PSD do not impair subsequent acquisition or short-term retention (1-hr) of a one-trial passive avoidance, but prevent long-term storage of this task. Linden *et al.* [28] maintain that PSD does not interfere with the initial learning but rather with the conversion of memory from the short-term to the long-term storage. This claim is supported by experiments showing that PSD in the acquisition-retention interval makes the memory trace susceptible to electroconvulsive shock disruption even days after learning [12,14] and that pre-acquisition PSD (3 days) protracts the retrograde amnesia gradient established by posttrial ECS [14,28]. Skinner *et al.* [37] reported that 3-day PSD preceding passive avoidance acquisition did not modify the behaviour of rats during learning but produced almost complete amnesia on the retention test 3 days later.

Although the present experiments are not directly relevant to the above controversy they suggest a possible means to dissociate the PSD effect on the initial CS processing and short-term memory formation from the

subsequent influences on short-term memory persistence and gustatory trace-poisoning association. Experiment 4 demonstrated that PSD impairs formation of gustatory trace, but absence of the latter prevented investigation of the PSD effect on the gustatory trace-poisoning association. Since CTA can be established even with extremely long (8 to 12 hr) CS-US intervals [32, 35, 36] it would be possible to compare the CTA impairing effects of PSD treatments of the same duration either preceding presentation of the gustatory CS or maintained throughout the prolonged CS-US interval. The 2-hr CS-US delay employed in Experiment 3 was obviously too short to allow meaningful comparisons of this type. It demonstrated, however, that the stress, invariably present in most PSD treatments, rather enhances than blocks CTA acquisition. The confinement stress affects CTA at a time when PSD is still ineffective; later it is probably masked by the stronger CTA impairing effects of PSD.

The PSD influence on retrieval is less controversial. According to Joy and Prinz [20], 4-day PSD before

retention testing of an active avoidance acquired 2 months earlier does not impair but rather improves retrieval. Similar results were obtained by Albert *et al.* [1] and Kitahama *et al.* [21]. On the other hand, the retention deficit observed by Leconte and Bloch [25] after 2-day PSD could be due to incomplete acquisition of the avoidance task at the PSD onset. Results of Experiment 5 indicate that CTA retrieval is not disrupted but rather enhanced by PSD.

It is not known how much free sleep is needed for removal of the behavioural consequences of PSD. Stern [39] reported that impairment of PAR acquisition by 5-day PSD is only partly relieved when 1 to 4 days of normal sleep intervene between PSD and learning. In this case behavioural compensation of the PSD effects proceeded more slowly than the electrophysiological one, since the REM sleep rebound completely disappears after three days [30]. Recovery of CTA in Experiment 2 seems to be more rapid probably because PSD was relatively short in this case. Further research should establish the relationship between PSD duration and recovery time in more detail.

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